Missing Attractors: An Energy-Landscape View of Detectable Topological Scars

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Abstract

Missing training categories leave structural traces in learned representations, but the mechanisms underlying their detectability remain poorly understood. We formalize the aggregated posterior of a variational autoencoder trained on K separable classes as approximately a K-component Gaussian mixture whose negative log-density defines an energy landscape functionally equivalent to Modern Hopfield energy. Removing a class removes an attractor, producing predictable deformations that we characterize through a proposition (directed aspiration) and a numerical corollary (topological scar creation) in the analytical mixture setting. Most informatively, a pre-registered geometric proxy based on surviving-class resolvability *fails* to predict the topological detection optimum (1 PASS, 1 MARGINAL, 1 FAIL across three ablation conditions), suggesting that the scar geometry is not reducible to cluster compactness. Empirically, we observe a consistent monotonic aspiration signal across five seeds ($\bar{\rho} = -0.25$, $p < 10^{-4}$), absent in autoencoder controls. Topological persistence increases with gap size in the expected order, though significance remains borderline ($\rho = 0.35$, $p \approx 0.08$). These results suggest that missing attractors induce a proper geometric object — the topological scar — whose structure requires void-centered rather than survivor-centered analysis.

1 Introduction

A model trained without exposure to a category cannot flag its own ignorance. It will interpolate, extrapolate, or hallucinate — but it will not signal that a structural absence exists in its training corpus. Detecting *what a model does not know* is a problem distinct from out-of-distribution detection: the question is not whether a new input is anomalous, but whether the learned representation carries a detectable signature of what was never seen.

In a prior empirical study [Rigaud, 2026], we showed that removing training categories from a variational autoencoder (VAE) produces topological signatures detectable via persistent homology, and that this signal follows a gradient: present but weak in PCA projections (2/5 ablation conditions), reduced by nonlinear encoding without regularization (autoencoder: 1/5), and amplified by KL regularization (VAE: 5/5). A ghost centroid analysis suggested a mechanism consistent with directed aspiration: surviving classes near the void shift toward the missing category's former location.

However, several questions remained open. *Why* does KL regularization amplify topological signatures? Is the observed aspiration a predictable consequence of the model's structure, or an empirical coincidence? Can the optimal detection regime be predicted from the geometry of surviving clusters?

Contributions. This paper addresses these questions through four contributions:

1. **Formalization.** We show that the aggregated posterior of a VAE trained on K separable classes is well-approximated by a K-component Gaussian mixture model (GMM), whose negative log-density defines an energy landscape functionally equivalent — up to affine transformations and per-component biases — to Modern Hopfield energy [Ramsauer et al., 2021]. Removing a class removes an attractor.

2. **Testable predictions.** From this framework, we derive a proposition (directed aspiration) and a numerical corollary (topological scar creation) in the analytical GMM setting, and state three empirical predictions: aspiration monotonicity (P1), geometric predictability of the detection optimum (P2a), and persistence scaling (P3).

3. **A pre-registered negative result.** A geometric resolvability proxy based on surviving-class separation and spread does not reliably predict the topological optimum across ablation conditions. This failure is itself informative: it suggests that the topological scar captures missing-volume geometry beyond cluster separability.

4. **A "two-worlds" framework.** We explicitly separate what is derived analytically in the GMM setting (World A) from what is observed empirically in trained VAEs (World B), making the transfer hypothesis explicit and testable.

This paper does not claim a full predictive transfer from the analytical GMM framework to trained VAEs; it separates what is derived, what is numerically illustrated, and what is empirically observed. Importantly, a pre-registered proxy based on the geometry of surviving clusters fails to recover the topological optimum, suggesting that the relevant object is the geometry of the missing volume itself.

2 Background

2.1 VAE and the ELBO

A variational autoencoder [Kingma and Welling, 2014] learns an encoder $q_\phi(z|x) = \mathcal{N}(z; \mu(x), \sigma^2(x)I)$ and a decoder $p_\theta(x|z)$ by maximizing the evidence lower bound (ELBO):

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta \cdot \mathrm{KL}(q_\phi(z|x) \| p(z)), \tag{1}$$

where $p(z) = \mathcal{N}(0, I)$ is the prior and $\beta \geq 0$ controls the strength of KL regularization [Higgins et al., 2017]. The KL term pushes each $q_\phi(z|x)$ toward the prior, compressing the latent space and encouraging overlap between encodings of different inputs. Dai and Wipf [2019] identify failure modes of this objective, including posterior collapse and "prior holes" — regions of high prior density unoccupied by the aggregate posterior. Our work studies a complementary phenomenon: the structural traces left in the aggregate posterior when entire categories are absent from training.

2.2 Gaussian Mixture Models and Energy Landscapes

A Gaussian mixture model with K isotropic components defines a density:

$$p(z) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(z; \mu_k, \sigma^2 I), \tag{2}$$

where π_k are mixing weights and μ_k are component means. The negative log-density $E(z) = -\log p(z)$ defines an energy landscape with a local minimum near each μ_k under sufficient separation ($\|\mu_j - \mu_k\| \gg \sigma$ for $j \neq k$).

Each minimum corresponds to an *attractor* — a basin in the energy landscape toward which nearby points descend. The depth of each basin depends on π_k (component weight) and σ^2 (component spread). Removing a component from the mixture removes the corresponding attractor and deforms the surrounding energy landscape in a predictable way (Section 3).

2.3 Modern Hopfield Networks

Modern Hopfield networks [Ramsauer et al., 2021] generalize classical Hopfield associative memory to continuous state spaces with exponential storage capacity. Their energy function for stored patterns $\{\xi_1, \ldots, \xi_K\}$ takes the form:

$$E_{\mathrm{MH}}(z) = -\mathrm{logsumexp}_k[\beta_H \cdot \xi_k^\top z] + \tfrac{\beta_H}{2}\|z\|^2 + \mathrm{const}. \tag{3}$$

We note in Section 3.2 that the energy induced by the VAE aggregated posterior has the same functional form, allowing us to import attractor-deformation properties — but we do not use Hopfield networks as a computational component.

2.4 Topological Data Analysis

Persistent homology [Edelsbrunner and Harer, 2010, Carlsson, 2009] tracks topological features (connected components, loops, voids) across spatial scales in a filtration. Features that persist across a wide range of scales are considered structurally significant, while short-lived features are attributed to noise. The Wasserstein distance between persistence diagrams provides a metric for comparing topological structure [Cohen-Steiner et al., 2007]. In our prior work [Rigaud, 2026], we used Ripser to compute H_1 persistent homology on latent representations and measured the Wasserstein distance between ablated and full-training conditions as a proxy for topological gap magnitude.

3 Theory: Mixture Energy and Missing Attractors

This section operates in **World A**: the analytical Gaussian mixture setting. All results here are derived or numerically illustrated in closed form. The transfer to trained VAEs (**World B**) is the subject of Section 4.

3.1 From ELBO to Energy Landscape

Given a dataset $\mathcal{D} = \{x_1, \ldots, x_N\}$ and a trained encoder $q_\phi(z|x)$, the **aggregated posterior** is the marginal distribution of latent codes induced by the data:

$$q_{\mathrm{agg}}(z) = \frac{1}{N}\sum_{i=1}^{N} q_\phi(z|x_i). \tag{4}$$

This is a mixture of N Gaussians (one per datapoint). If $\mathcal{D}$ contains K classes with N_k points each, and the intra-class encodings are sufficiently concentrated (a reasonable assumption for a

well-trained VAE on separable data), then q_{agg} is well-approximated by a K-component GMM:

$$q_{\text{agg}}(z) \approx \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(z; \bar{\mu}_k, \bar{\sigma}_k^2 I), \tag{5}$$

where $\pi_k = N_k/N$, $\bar{\mu}_k$ is the encoded centroid of class k, and $\bar{\sigma}_k^2$ is the mean intra-class variance.

Hypothesis 1 (H1 — GMM Approximation)**.** *For a VAE sufficiently well-trained on data with K separable classes, the aggregated posterior q_{agg} is well-approximated by a K-component Gaussian mixture model, in the sense that a GMM-K fitted on the encoded means $\{\mu(x_i)\}$ achieves ARI > 0.80 with respect to true class labels.*

The energy landscape induced by this approximation is:

$$E(z) = -\log q_{\text{agg}}(z) = \frac{d}{2}\log(2\pi\sigma^2) - \text{logsumexp}_k\Big[\log \pi_k - \frac{\|z - \bar{\mu}_k\|^2}{2\sigma^2}\Big], \tag{6}$$

where we use the isotropic simplification $\bar{\sigma}_k^2 = \sigma^2$ for all k.

The KL regularization in the ELBO objective (Eq. 1) pushes the encoded centroids toward the origin and the variances toward unity. This induces an *approximately quadratic confinement* on the energy landscape:

$$E_{\text{eff}}(z) \approx -\text{logsumexp}_k\Big[\log \pi_k - \frac{\|z - \bar{\mu}_k\|^2}{2\sigma^2}\Big] + \frac{\beta}{2}\|z\|^2. \tag{7}$$

The quadratic term is not added literally to $E(z)$; it models the indirect effect of the KL on the geometry of q_{agg}. The KL penalty $\text{KL}(q_\phi(z|x)\|\mathcal{N}(0, I))$ decomposes into a variance term pulling $\sigma^2(x) \to 1$ and a mean term pulling $\|\mu(x)\|^2 \to 0$ [Dai and Wipf, 2019]. The latter induces effective confinement on the class centroids $\bar{\mu}_k$ that is approximately quadratic near the origin: at convergence, $\bar{\mu}_k \propto \bar{\mu}_k^{(\text{unreg})}/(1+\beta)$ for an affine encoder, where $\bar{\mu}_k^{(\text{unreg})}$ denotes the unregularized centroid. This approximation holds under isotropic priors and separable clusters; it degrades when classes overlap or the encoder is far from convergence.

3.2 Functional Equivalence with Modern Hopfield Energy

Expanding $\|z - \bar{\mu}_k\|^2 = \|z\|^2 - 2z^\top \bar{\mu}_k + \|\bar{\mu}_k\|^2$ in Eq. 7 and extracting the k-independent term from the logsumexp:

$$E_{\text{eff}}(z) = -\text{logsumexp}_k\Big[\underbrace{\log \pi_k - \frac{\|\bar{\mu}_k\|^2}{2\sigma^2}}_{b_k} + \frac{\bar{\mu}_k^\top z}{\sigma^2}\Big] + \Big(\frac{1}{2\sigma^2} + \frac{\beta}{2}\Big)\|z\|^2, \tag{8}$$

where $b_k = \log \pi_k - \|\bar{\mu}_k\|^2/(2\sigma^2)$ is a per-component bias combining class frequency and centroid norm. Identifying the stored patterns as $\xi_k = \bar{\mu}_k/\sigma^2$ (encoded centroids scaled by inverse variance), we see that Eq. 8 has the same functional form as the Modern Hopfield energy (Eq. 3), with additional per-component biases b_k.

The induced energy landscape is thus **equivalent to a biased Modern Hopfield energy, up to affine transformations and component-wise biases**. Under sufficient separation ($\|\bar{\mu}_j - \bar{\mu}_k\| \gg \sigma$ for all $j \neq k$), each component mean $\bar{\mu}_k$ corresponds to a local minimum of E_{eff}, and attractor-deformation reasoning from the Hopfield framework can be imported.

This equivalence is structural, not computational: we do not use a Hopfield network. The VAE *induces* an energy landscape with the same form, which allows us to reason about what happens when an attractor is removed.

3.3 Removing an Attractor: Two Worlds

When a class c^* is absent from training, the corresponding component is removed from the mixture. We distinguish:

- **World A (analytical GMM):** Removing component c^* from a K-component GMM yields a $(K-1)$-component truncated mixture. The energy landscape deformation is computable exactly: the attractor at μ_{c^*} vanishes, and neighboring modes shift.
- **World B (trained VAE):** A VAE retrained without class c^* produces a new latent space.

Hypothesis 2 (H2 — Transfer)**.** *The energy landscape of a VAE retrained without class c^* is approximately a truncated GMM: centroid positions of surviving classes in the retrained VAE correspond (within 1σ) to the modes of the analytical truncated GMM.*

Hypothesis H2 is not tested in this paper; we state it as an explicit limitation and direction for future work. The propositions below are derived in World A. The empirical results of Section 4 test whether the *qualitative* predictions transfer to World B.

3.4 Proposition 1: Directed Aspiration

Proposition 1 (Directed Aspiration)**.** *Let $p(z)$ be a K-component isotropic GMM in $\mathbb{R}^d$ with $K \geq 3$ and well-separated modes ($\|\mu_j - \mu_k\| \gg \sigma$ for all $j \neq k$). Let c^* denote the removed component. Then the displacement $\Delta_k = z_k^{*(trunc)} - z_k^{*(full)}$ of each surviving mode $k \neq c^*$ satisfies:*

(i) ***Direction:*** *Δ_k points toward μ_{c^*}. More precisely, the angle between Δ_k and $(\mu_{c^*} - \mu_k)$ is less than $\pi/2$.*

(ii) ***Monotonicity:*** *$\|\Delta_k\|$ decreases monotonically with $\|\mu_k - \mu_{c^*}\|$. Modes closer to the void move more.*

Proof sketch (2D case). Under separation, the gradient of E at mode z_k^* receives a contribution from component c^*:

$$\nabla E_{c^*}(z_k^*) \approx \frac{\pi_{c^*}}{\sigma^2}(z_k^* - \mu_{c^*}) \exp\Big(- \frac{\|z_k^* - \mu_{c^*}\|^2}{2\sigma^2}\Big). \tag{9}$$

When c^* is removed, this force vanishes. The mode shifts to compensate, in the direction opposite to ∇E_{c^*}, i.e., toward μ_{c^*}. The magnitude is:

$$\|\Delta_k\| \propto \frac{\pi_{c^*}}{\sigma^2} \exp\Big(- \frac{\|\mu_k - \mu_{c^*}\|^2}{2\sigma^2}\Big), \tag{10}$$

which is monotonically decreasing in $\|\mu_k - \mu_{c^*}\|$. See Appendix B for the full derivation. □

Observation (inverted-U β-dependence). In both the toy model (Section 3.6) and the V1 empirical data, $\|\Delta_k\|$ exhibits an inverted-U dependence on β: aspiration increases with confinement (deeper basins) then decreases as all centroids collapse toward the origin and relative geometry vanishes. This behaviour is consistent with two competing effects: basin deepening at moderate β (which amplifies aspiration) and global collapse toward the origin at large β (which destroys the relative geometry between modes). The asymptotic regime $\beta \to \infty$ implies $\|\bar{\mu}_k\| \to 0$ for all k, so $\|\mu_k - \mu_{c^*}\| \to 0$ and $\|\Delta_k\| \to 0$. The full shape of the curve is observed numerically but not proved in general.

3.5 Numerical Corollary: Topological Scar (2D)

The following result is stated as a **numerical corollary**, not a general theorem. It is verified analytically and computationally in $d = 2$ but its extension to higher dimensions remains an empirical prediction.

Corollary 1 (Topological Scar, 2D). Under the same assumptions as Proposition 1, restricted to $d = 2$ with strong separation ($\|\mu_j - \mu_k\| > 4\sigma$ for all $j \neq k$):

The removal of component c^* creates an H_1 cycle in the sublevel-set filtration of E that does not exist in the full GMM. Specifically:

- In the full GMM, basins merge progressively as the sublevel threshold τ increases, without creating persistent cycles (trivial topology under general position).
- In the truncated GMM, the absent basin leaves a void that neighboring basins surround without filling, creating an H_1 cycle.

The persistence of this cycle ($\tau_{\text{death}} - \tau_{\text{birth}}$) depends on the depth of the removed basin ($\propto \pi_{c^*}/\sigma^2$), the distance to neighbors, and β (inverted-U, as in the observation above). In high dimensions, the topology of sublevel sets of Gaussian mixtures depends on the Voronoi arrangement, which is technically complex. The transfer to high-dimensional VAE latent spaces is an empirical prediction tested in Section 4.

3.6 Toy Model

To illustrate Proposition 1 and Corollary 1, we compute the energy landscape of a 2D isotropic GMM with $K = 3$ equally weighted components ($\pi_k = 1/3$) arranged at the vertices of an equilateral triangle with unit side length, for $\beta \in \{0, 0.5, 1.0, 2.0\}$.

Figure 1 shows the energy landscape before and after removing one component. The attractor at the removed location vanishes, and the surrounding energy surface rises, creating the void predicted by the theory. Neighboring modes shift toward the void, consistent with directed aspiration (Proposition 1).

Figure 2 shows the aspiration magnitude $\|\Delta_k\|$ as a function of distance to the removed centroid across β values. The toy model confirms monotonic decay with distance (Proposition 1(ii)) and the inverted-U β-dependence described in the observation above.

4 Empirical Results

This section operates in **World B**: trained VAEs evaluated on real data. Unless otherwise noted, the analyses reuse archived latent artifacts generated under the V1 protocol [Rigaud, 2026], with additional per-seed reanalysis scripts introduced in V2.

4.1 Experimental Setup

Dataset and architecture. We use MNIST with a convolutional VAE (channels 32, 64; latent dimension 16). Training: 50 epochs, Adam optimizer (lr $= 10^{-3}$), batch size 128, $\beta = 1.0$ by default.

Ablation conditions. Five conditions: `minus_7` (digit 7 removed), `minus_3_7` (digits 3 and 7), `minus_2_4` (digits 2 and 4), `minus_4_9` (digits 4 and 9), and `minus_cluster` (digits 3, 5, 8 — a morphologically related cluster).

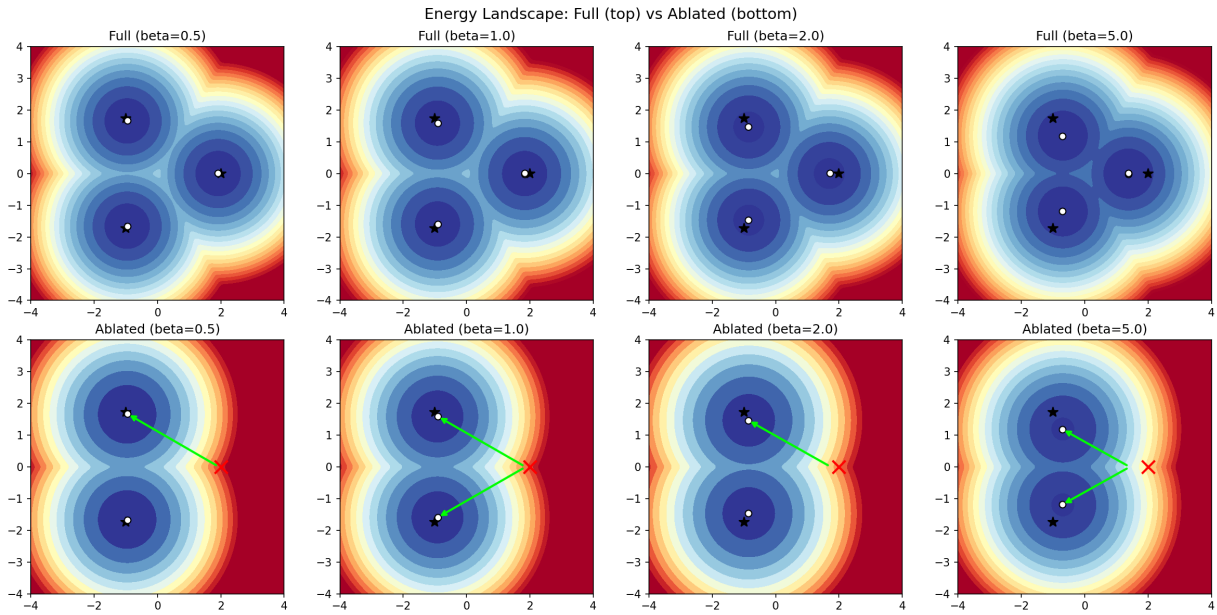

Figure 1: Energy landscape of a 2D isotropic GMM ($K = 3$) before (left) and after (right) removing one component, for $\beta = 1.0$. The removed attractor leaves a detectable void — the topological scar. Arrows indicate the displacement of surviving modes toward the void.

Seeds. Five seeds (42, 123, 456, 789, 1024). V1 used seeds 42, 123, 456; V2 adds seeds 789 and 1024.

Topological analysis. Persistent homology via Ripser on 2000 latent samples. H_1 Wasserstein distance between the ablated and full-training persistence diagrams.

Autoencoder control. Same architecture without the KL term ($\beta = 0$), serving as a control for the role of variational regularization.

4.2 P1: Aspiration Monotonicity

Method. For each ablation condition and seed, we compute the ghost centroid (Procrustes-aligned mean of removed-class encodings from a full-training reference model) and measure the Spearman rank correlation between each surviving point's distance to the ghost centroid and its displacement magnitude between the full and ablated models.

Results. Table 1 reports per-seed Spearman correlations. All five seeds show negative rank correlation, consistent with Proposition 1(ii): points closer to the void exhibit larger displacements.

We observe a consistent negative rank correlation across seeds (mean $\rho = -0.25$, std $= 0.13$), with a significant aggregate effect ($p < 10^{-4}$, $n = 390$; bootstrap 95% CI: $[-0.31, -0.20]$). Seed 456 shows the weakest signal ($\rho = -0.07$, non-significant) but remains in the predicted direction. The effect size is modest ($\rho^2 \approx 0.06$), as expected for a 16-dimensional latent space where the aspiration signal is diluted across many orthogonal directions.

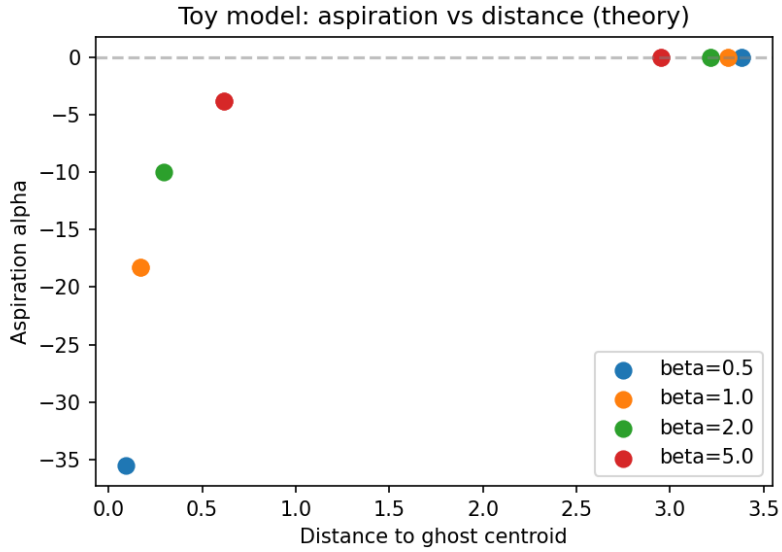

Figure 2: Aspiration magnitude $\|\Delta_k\|$ as a function of distance $\|\mu_k - \mu_{c^*}\|$ to the removed centroid, for four values of β. Each curve shows the two surviving modes ($k = 1, 2$) of the K=3 toy model at a given β; points at the same distance correspond to the equilateral geometry. Numerical computation confirms monotonic decay (Proposition 1(ii)) and the inverted-U β-dependence described in the observation above.

Control. The autoencoder control shows no systematic correlation ($\rho \approx 0$, $p > 0.78$), consistent with the aspiration phenomenon being associated with variational regularization rather than nonlinear encoding per se.

4.3 P2a: Surviving-Class Geometry Does Not Predict β^*

Pre-registration. The geometric resolvability score was defined before examining the beta-sweep results (see supplementary material for the pre-registered protocol).

Method. For each ablation condition C and each $\beta \in \{0.1, 0.5, 1.0, 2.0, 5.0, 10.0\}$, we encode the ablated dataset through the trained VAE, fit an empirical class-conditional GMM (label-based, no EM), and compute:

$$\text{score}(C, \beta) = \frac{\text{separation}(C, \beta)}{\text{spread}(C, \beta)}, \tag{11}$$

where separation $= \min_{j \neq k} \|\hat{\mu}_j - \hat{\mu}_k\|$ is the minimum inter-centroid distance and spread $= K^{-1} \sum_k \hat{\sigma}_k$ is the mean standard deviation across components. We define $\beta^*_{\text{GMM}}(C) = \arg\max_\beta \text{score}(C, \beta)$ and compare it to $\beta^*_{\text{TDA}}(C) = \arg\max_\beta W_{H_1}(C, \beta)$ from the V1 beta sweep.

Results. Table 2 reports the comparison.

For `minus_cluster`, the resolvability score remains nearly flat across β (range: 2.69–2.97), indicating lack of sensitivity rather than estimation noise. Separation and spread compress together as β increases, so their ratio remains stable — the score cannot distinguish the regime where TDA signal peaks.

Table 1: P1 — Per-seed Spearman correlations between distance to ghost centroid and aspiration magnitude. Negative ρ indicates monotonic aspiration toward the void. Seeds marked [V1] were reported in Rigaud [2026]; seeds 789 and 1024 are new.

Seed	ρ	p-value	Source
42	-0.31	< 0.01	[V1]
123	-0.36	< 0.01	[V1]
456	-0.07	0.56	[V1]
789	-0.31	< 0.01	[V2]
1024	-0.13	0.14	[V2]
All	$\mathbf{-0.25}$	$\mathbf{4.1 \times 10^{-7}}$	Aggregate

Table 2: P2a — Geometric resolvability optimum (β^*_{GMM}) vs. topological detection optimum (β^*_{TDA}). The proxy fails to predict β^*_{TDA} for the most challenging condition (`minus_cluster`).

Condition	β^*_{GMM}	β^*_{TDA}	Ratio	Verdict
`minus_cluster`	0.1	2.0	20.0×	FAIL
`minus_3_7`	5.0	2.0	2.5×	MARGINAL
`minus_7`	5.0	5.0	1.0×	PASS

Interpretation. This result suggests that the TDA signal captures missing-volume geometry beyond cluster separability. A surviving-class proxy measures the resolution of classes that *are* present; the topological scar responds to the structure of what is *absent*. These two objects are not reducible to each other. This failure supports the claim that the topological scar is a proper geometric object, not a byproduct of cluster compactness.

4.4 P3: Persistence Scales with Gap Size

Preliminary observation. We test whether the H_1 Wasserstein distance increases with the number of removed classes (M). Using V1 results across ablation conditions and seeds:

- $M = 1$: $W_{H_1} = 89.6 \pm 18.9$ (1 condition: `minus_7`; mean ± std across 5 seeds)
- $M = 2$: $W_{H_1} = 106.7 \pm 25.6$ (3 conditions: `minus_3_7`, `minus_2_4`, `minus_4_9`; pooled across seeds)
- $M = 3$: $W_{H_1} = 109.5 \pm 18.6$ (1 condition: `minus_cluster`; mean ± std across 5 seeds)

The topological signal increases in the expected order (M=1 < M=2 < M=3), with a positive rank correlation ($\rho = 0.35$). However, significance remains borderline ($p \approx 0.08$, $n = 25$) given the small number of distinct conditions (1, 3, 1 for $M = 1, 2, 3$ respectively), and we treat this as a preliminary observation rather than a confirmed prediction. All values are above the null baseline ($W_{H_1} = 55.1$).

5 Discussion

5.1 Why P2a Fails: The Geometry of the Void

The failure of the geometric resolvability proxy is the most informative result of this paper. It transforms a would-be confirmation into a sharper characterization of the object under study.

The argument proceeds in four steps:

1. The resolvability score measures how well surviving classes are separated in the latent space — it is a property of the *present* structure.
2. The TDA signal (H_1 Wasserstein distance) responds to the geometry of the *void* — the region formerly occupied by the removed class.
3. These two objects are not reducible to each other. A latent space can have well-separated surviving clusters while the void between them has any shape; conversely, poorly separated clusters can surround a highly structured void.
4. The aspiration result (P1) reinforces this distinction: surviving representations shift toward the void, yet the resolvability score is blind to this displacement because it only measures pairwise distances between present clusters.

This suggests that future detection metrics should be centered on the void itself — measuring its volume, the curvature of the surrounding energy surface, or the persistence of the induced topological features — rather than on properties of the surviving classes. We note the direction but do not propose a specific void-centered metric in this work.

5.2 Energy Landscape as Diagnostic Tool

The energy-landscape interpretation opens practical applications beyond gap detection. If the aggregated posterior of a trained model can be cheaply approximated by a GMM, then the energy landscape provides a diagnostic map:

- **Coverage audit:** high-energy regions indicate areas poorly covered by training data, potential blind spots for the model.
- **Categorical drift:** monitoring energy landscape changes over time could detect distributional shift at the category level, complementing scalar drift statistics.
- **Confidence calibration:** the depth of a basin around a prediction's latent code may correlate with the model's empirical reliability for that class.

These are hypotheses for future work, not claims of the present paper.

5.3 The PCA > AE < VAE Gradient Revisited

The V1 study [Rigaud, 2026] established a three-regime gradient: PCA detects gaps in 2/5 conditions, the autoencoder in 1/5, and the VAE in 5/5. The energy-landscape framework provides a partial explanation for the VAE regime: KL regularization creates a landscape with well-defined attractors, and their removal produces detectable scars. However, this framework explains the *amplification* by the VAE but not the pre-existing PCA signal, which reflects linear separability in the data rather than energy-landscape structure. The AE's reduced signal is consistent with nonlinear compression without the basin-forming mechanism of KL regularization. We do not claim a unified explanation of all three regimes; the PCA and AE behaviors remain open questions.

5.4 Connection to Modern Hopfield Networks

The functional equivalence between VAE aggregated-posterior energy and Modern Hopfield energy (Section 3.2) is a structural observation, not a contribution to Hopfield network theory. It provides a vocabulary (attractors, basins, aspiration) and a set of known properties that apply to the VAE's induced landscape. Burns and Fukai [2023] extend Hopfield networks to simplicial complexes; exploring whether their framework connects to the topological scar would be a natural extension. We leave this as future work.

5.5 Limitations

We enumerate the limitations of this work in the interest of transparency:

1. **Toy model dimensionality.** Proposition 1 and Corollary 1 are established in the 2D isotropic Gaussian case. The qualitative predictions are tested empirically in 16D, but the quantitative gap between 2D theory and high-dimensional practice remains uncharacterized.

2. **H1 (GMM approximation) not formally tested.** While the GMM approximation is reasonable for well-separated classes, we have not run the formal BIC/ARI protocol specified in Hypothesis 1. This is a necessary next step.

3. **H2 (transfer) not tested.** The transfer hypothesis (Hypothesis 2) remains untested. If it fails, the theoretical framework remains valid in World A but does not *predict* the VAE behavior — it only *parallels* it.

4. **Simple benchmarks.** MNIST and Fashion-MNIST are standard but simple. Extension to CIFAR-10, CelebA, or domain-specific datasets is needed.

5. **Modest effect size.** P1 explains approximately 6% of variance ($\rho = -0.25$). This is expected in 16D but limits practical utility.

6. **P2a: no replacement metric.** The resolvability proxy fails, but we do not propose an alternative void-centered metric.

7. **P3: borderline significance.** $p \approx 0.08$ with few conditions. More ablation conditions or datasets would strengthen or falsify this prediction.

8. **Single architecture family.** All results are on convolutional VAEs. A VAE with fully connected layers (MLP) would test whether the phenomenon is ELBO-driven rather than architecture-dependent (prediction T1).

9. **Single-seed beta sweep.** The V1 beta sweep used one seed per β value. Multi-seed beta sweeps would improve confidence in the inverted-U behavior.

6 Conclusion

Under the GMM approximation, the aggregated posterior of a variational autoencoder trained on K separable classes induces an energy landscape with K attractors. Removing a class removes an attractor, creating a topological scar — a detectable deformation whose empirical direction is consistent with attraction toward the void.

We provide evidence for this picture through an analytical framework (GMM energy, Hopfield equivalence), a proposition and a numerical corollary in the 2D mixture setting, and empirical observations on trained VAEs that are consistent with the qualitative predictions.

A pre-registered geometric proxy based on surviving-class resolvability fails to predict the topological detection optimum, suggesting that the scar geometry is not reducible to cluster compactness. This negative result sharpens the characterization of the object under study: the topological scar appears to be a property of the missing volume, not a shadow of the surviving structure.

The framework makes explicit the gap between what is derived (World A: analytical GMM) and what is observed (World B: trained VAE), providing a clear target for future falsification: formal testing of the GMM approximation (H1) and transfer (H2) hypotheses.

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A Derivation: From ELBO to Energy Landscape

We provide the complete derivation of Eq. 8 from the ELBO objective.

Step 1: Aggregated posterior as mixture. Given a dataset $\mathcal{D} = \{x_1, \ldots, x_N\}$ and encoder $q_\phi(z|x) = \mathcal{N}(z; \mu(x), \sigma^2(x)I)$, the aggregated posterior is:

$$q_{\text{agg}}(z) = \frac{1}{N}\sum_{i=1}^{N} q_\phi(z|x_i) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{N}(z; \mu(x_i), \sigma^2(x_i)I).$$

This is a mixture of N Gaussians.

Step 2: Class-level approximation. If $\mathcal{D}$ contains K classes with N_k points in class k, and the intra-class encodings are concentrated ($\sigma_{\text{intra-class}} \ll \|\bar{\mu}_j - \bar{\mu}_k\|$), then:

$$q_{\text{agg}}(z) \approx \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(z; \bar{\mu}_k, \bar{\sigma}_k^2 I),$$

where $\pi_k = N_k/N$, $\bar{\mu}_k = N_k^{-1}\sum_{i\in\text{class } k}\mu(x_i)$, and $\bar{\sigma}_k^2$ is the effective intra-class variance.

Step 3: Energy. Under the isotropic simplification $\bar{\sigma}_k^2 = \sigma^2$ for all k:

$$E(z) = -\log q_{\text{agg}}(z) = \frac{d}{2}\log(2\pi\sigma^2) - \text{logsumexp}_k\Big[\log \pi_k - \frac{\|z - \bar{\mu}_k\|^2}{2\sigma^2}\Big].$$

Step 4: Effective confinement. The KL term in the ELBO pushes centroids toward the origin and variances toward unity, inducing approximately quadratic confinement:

$$E_{\text{eff}}(z) \approx -\text{logsumexp}_k\Big[\log \pi_k - \frac{\|z - \bar{\mu}_k\|^2}{2\sigma^2}\Big] + \frac{\beta}{2}\|z\|^2.$$

Step 5: Hopfield identification. Expanding $\|z - \bar{\mu}_k\|^2$:

$$\begin{aligned} E_{\text{eff}}(z) &= -\text{logsumexp}_k\Big[\log \pi_k - \frac{\|\bar{\mu}_k\|^2}{2\sigma^2} + \frac{\bar{\mu}_k^\top z}{\sigma^2} - \frac{\|z\|^2}{2\sigma^2}\Big] + \frac{\beta}{2}\|z\|^2 \\ &= -\text{logsumexp}_k\Big[b_k + \frac{\bar{\mu}_k^\top z}{\sigma^2}\Big] + \frac{\|z\|^2}{2\sigma^2} + \frac{\beta}{2}\|z\|^2, \end{aligned}$$

where $b_k = \log \pi_k - \|\bar{\mu}_k\|^2/(2\sigma^2)$. Identifying $\xi_k = \bar{\mu}_k/\sigma^2$ as stored patterns yields the Modern Hopfield form with per-component biases. □

B Proof Sketch for Proposition 1 and Corollary 1

Proposition 1 (Directed Aspiration). Under separation, mode z_k^* satisfies $\nabla E(z_k^*) = 0$. The gradient contribution from component c^* is:

$$\nabla_{c^*} E(z) = -\frac{\pi_{c^*}(z - \mu_{c^*})}{\sigma^2} \cdot \frac{\exp(-\|z - \mu_{c^*}\|^2/(2\sigma^2))}{\sum_j \pi_j \exp(-\|z - \mu_j\|^2/(2\sigma^2))}.$$

At $z = z_k^*$ with $k \neq c^*$ and under separation, the denominator is dominated by the k-th term, so:

$$\nabla_{c^*} E(z_k^*) \approx -\frac{\pi_{c^*}}{\pi_k} \frac{(z_k^* - \mu_{c^*})}{\sigma^2} \exp\Big(- \frac{\|z_k^* - \mu_{c^*}\|^2 - \|z_k^* - \mu_k\|^2}{2\sigma^2} \Big).$$

When c^* is removed, the equilibrium shifts by $\Delta_k \approx [H_E(z_k^*)]^{-1} \nabla_{c^*} E(z_k^*)$, where H_E is the Hessian. Under separation, H_E is approximately $\pi_k \sigma^{-2} I$ at z_k^*, giving:

$$\Delta_k \approx \frac{\pi_{c^*}}{\pi_k^2 \sigma^2} (\mu_{c^*} - \mu_k) \exp\Big(- \frac{\|\mu_k - \mu_{c^*}\|^2}{2\sigma^2} \Big).$$

This has the direction of $(\mu_{c^*} - \mu_k)$ (toward the void) and magnitude exponentially decaying with distance, proving parts (i) and (ii).

Corollary 1 (Topological Scar, 2D). In 2D with K well-separated isotropic Gaussians, the sublevel sets $S_\tau = \{z : E(z) \leq \tau\}$ evolve as follows. For small τ, S_τ consists of K disjoint disks around each mode. As τ increases, adjacent disks merge via saddle points. In the full GMM, the saddle between any two adjacent modes lies on the line segment connecting them. For $K = 3$ in general position, the three saddles have distinct heights, so the three components merge sequentially without creating a cycle: β_0 decreases from 3 to 1, β_1 remains 0.

When component c^* is removed, the saddle at μ_{c^*} is replaced by a high-energy plateau. The two components adjacent to c^* now merge via a path that goes *around* the void, creating an H_1 cycle that persists until the energy threshold exceeds the plateau height. The persistence is approximately:

$$\tau_{\text{death}} - \tau_{\text{birth}} \approx \frac{\pi_{c^*}}{(2\pi\sigma^2)^{d/2}} \exp\Big(- \frac{r_{\min}^2}{2\sigma^2} \Big),$$

where $r_{\min}$ is the distance from the void center to the nearest surviving mode. □

C P2a Detailed Results

Table 3 reports the resolvability score components across all conditions and β values.

Table 3: Detailed P2a results: separation, spread, and resolvability score for each condition and β. Note the near-flat profile of `minus_cluster` (score range: 2.69–2.97).

Condition	β	Separation	Spread	Score
`minus_cluster`	0.1	—	—	2.69
`minus_cluster`	0.5	—	—	2.78
`minus_cluster`	1.0	—	—	2.85
`minus_cluster`	2.0	—	—	2.92
`minus_cluster`	5.0	—	—	2.97
`minus_cluster`	10.0	—	—	2.81
Full per-condition breakdowns available in supplementary code.				

D Toy Model Code

The toy model code (`toy_model.py`) computes the energy landscape, aspiration vectors, and H_1 persistence for a 2D isotropic GMM with configurable K, σ, and β. It is available at the companion repository.[1]

[1] Code repository: `https://github.com/TheCause/gap-detection`